Daniel Hardesty Lewis

Core Expertise

8 years **ML Engineering**: Build and operate AI/ML training and inference pipelines, with evaluation harnesses that catch regressions and improve production reliability

8 years **Backend & Data**: Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS

8 years **Distributed Systems**: Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency/throughput with explicit cost/perf tradeoffs

10 years **Geospatial AI**: Build spatial-temporal (geospatial + time-series) pipelines and models for risk, planning, and hazard decision-support use cases

8 years **Technical Leadership**: Drive execution through specs and stakeholder alignment; mentor engineers through reviews and instruction

Technologies

Modeling	PyTorch; TensorFlow; scikit-learn; Hugging Face Transformers; LoRA/QLoRA/PEFT
LLM apps	embeddings; RAG; LangChain; LangGraph; DSPy; FAISS
Data stack	Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; Redis; Elasticsearch; Supabase
Infra	Linux/Unix; Bash; Git; Docker; Kubernetes; Ray; Spark; Dask; Airflow; CI/CD; monitoring/observability
Additional	CUDA; C++; TypeScript; React

Experience

Professional

Apr. 2025 – Present **Co-founder**, *Properlytic*, New York, NY

Building a consumer real-estate pricing product focused on individual home price forecasting.

- Deployed a current-price prediction model for Manhattan condos and co-ops; achieved 12% MAPE on a holdout set and matched Zillow at 8.4% MAPE under an internal apples-to-apples evaluation protocol.
- Built a VAE-based latent representation layer over property, transaction, and neighborhood signals to stabilize learning and improve generalization under sparse and noisy features.
- Prototyped a diffusion-based forecasting backbone in Houston; measured 8% annualized compounding error on backtests spanning the 2020–2021 COVID period.
- Migrating market-specific models onto the diffusion backbone to standardize training, calibration, and cross-market evaluation.
- Conducted investor discussions with a16z, MetaProp, and other firms; refined product positioning and execution milestones from feedback.
- Pitch deck: link

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and data contracts for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, Pydantic-validated JSON interfaces, and FastAPI serving backed by Supabase and Redis
- Built the tariff simulation and what-if framework for policy scenarios affecting BOM cost, lead times, and compliance risk, including China decoupling and Section 301 exposure analysis
- Presented the prototype to enterprise stakeholders including KPMG; market validation: KPMG subsequently launched a tariff-modeling product
- Raising seed; running customer discovery with importers and exporters; iterating on distribution, pricing, and compliance operations

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows.

- Built a 0.5 m statewide seamless Texas DEM mosaic using nearest-neighbor stitching across 70+ state, federal, and local elevation sources.
- Built the web distribution platform end-to-end (data pipeline, tiling, hosting, delivery UX) to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

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Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP
- Led the source-to-frontend data pipeline for $40M Texas Disaster Information System (TDIS) disaster resiliency initiative
- Delivered summitgeospatial.com, a precision decision-support platform used by civil engineers to quantify risk and allocate resources for disaster planning
- Contributed research supporting Paola Passalacqua's 2022 EGU Bagnold Medal, among geomorphology's highest honors
- Scaled climate and flood models on world-leading supercomputers, executing million-node distributed jobs and managing multi-million-dollar compute budgets
- Evaluated and approved multi-year compute requests totaling tens of millions of dollars
- Managed relationships and developed projects with federal, state agencies, and universities including NOAA, NSF, US Army Corps of Engineers, Texas General Land Office, USC, UT, A&M

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers program, focusing on disaster resiliency in East Africa
- Research received a mention from President of AAAI Yolanda Gil during her farewell speech
- Helped teach Petrobras deep learning on HPC

Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
Targetting theory publications to DeepSeek's GRPO algorithm

Jan. 2025 – Present **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY
Achieved 85% R^2 relative to commercial SaaS (75%) and Fama-French (58%) on backtested holdout prediction of daily mutual fund returns

- Using a DL latent factor model to produce better factors than our benchmark: Fama-French's 5-factor model
- Deriving a tractable, axiom-compliant Per-Feature SHAP algorithm for deep learning using Hessian-based recursive clustering to stabilize attributions across temporal regimes

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into the MINT Platform ("DARPA-funded Airflow for science")

Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX
Helped teach graduate-level courses: *Machine Learning for the Geosciences*, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization

Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

Key Projects

oppsAlert — Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15000+ points of individual homeowner protest data provided by the City of Austin

Stealth, RegTech — *Seeking PE investment:* Teaming up with a top-proptech operator for a very large opportunity to aid cities achieve permitting compliance via simple pipeline collating publicly available imagery against permitting data

Flood Hazard Assessment — Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.

Texas Disaster Information System (TDIS) — Led development of the core elevation data layer for a web-based spatial data system that supports resilient decision-making at the state and local levels.

Real-Time Inundation Mapping — Developed computationally efficient methods to produce high-resolution (1m) flood inundation maps from National Water Model outputs for the Texas Division of Emergency Management (TDEM).

Languages

Spanish — Native / Bilingual Proficiency

French — Professional Working Proficiency

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